⛈️ The Strike & Staying Safe: Guided Notes · ANSWER KEY

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

Use the class slides and your reading. Key words:

potential energy | *thermal energy* | *stepped leader* | *return stroke* | *conductor* | *insulator* | *ion* | *current* | *dissolve* | *lightning rod*

A. Energy flow (L1.10)

1. Separated charge stores **potential** energy.

2. List the three forms of energy that potential energy changes into during a strike.
Thermal (heat), light, and sound.

B. The strike (L1.11)

3. Fill in the table.

Step	What happens
1. Stepped leader	**Negative charge zig-zags down from the cloud**
2. Streamer	**Positive charge reaches up from a tall object**
3. Return stroke	**They connect; huge current, bright flash**
4. Thunder	**Air heats and expands fast: boom**

C. Conductors and insulators (L1.12, L1.13.5)

4. In a **conductor**, electrons move easily. In an **insulator**, they do not.

5. Why are metals good conductors? Use the word electrons.
Some of their electrons are free to move from atom to atom.

6. Fill in the table.

Material	Conductor or insulator?
copper wire	**conductor**
rubber	**insulator**
salt water	**conductor**
dry wood	**insulator**

D. Salt water and safety (L1.13, L1.14)

7. **Reading the No Swimming data: 1 amp (A) = 6.28×10^{18} charges per second.**

> Example: 2 A = $2 \times 6.28 \times 10^{18} = 12.56 \times 10^{18} = 1.256 \times 10^{19}$ charges per second.

a) 0.5 A = ____ $\times 10^{18}$ charges per second **3.14**

b) 3 A = ____ $\times 10^{18}$ charges per second **18.84 (= 1.884×10^{19})**

8. Claim: How does dissolving salt in water affect the current? Give evidence from the lab and reasoning.
More salt → more current (brighter bulb, higher amps) because dissolved salt makes more ions that carry charge.

9. Draw a house with a lightning rod. Use arrows to show the path of the charge.
Rod on the roof, wire down the side, arrows from cloud → rod → wire → ground.

🔑 *Key written by our team from Mr. Brittle's worksheets and slides, please check.*